Nakaya's conjectures on the linear factors and degree of Fricke supersingular polynomials

David Jordan

Abstract

We deduce two conjectures of Nakaya on the supersingular polynomials $ss_p^{(N*)}$ attached to the Fricke groups $\Gamma_0(N)^+$, at all fourteen odd prime levels N in his set: the class-number formula for the number of linear factors over $\mathbb{F}_p$ (Conjecture 5) and the formula for the degree (Conjecture 6). Both follow from the Deligne–Rapoport model of $X_0(Np)$ modulo p, in which the Atkin–Lehner involution w_p acts as Frobenius on the supersingular points. Comparing genera turns the rational and the total supersingular counts on $X_0(N)^+$ into fixed-point counts of Atkin–Lehner involutions and genera of Atkin–Lehner quotients of $X_0(Np)$, which classical formulas evaluate. Together with Aricheta's genus formula, which shows that the remaining factors are $g\big(X_0(Np)/\langle w_N, w_p\rangle\big)$ irreducible quadratics, this determines the factorization of $ss_p^{(N*)}$ over $\mathbb{F}_p$. The same arguments count the rational and the total supersingular points on $X_0(N)^+$ at every odd prime level, the latter with a correction proportional to $g\big(X_0(N)^+\big)$. Exact computations, including direct counts from modular polynomials, are provided.

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1 Introduction

Deuring's class-number formulas relate supersingular elliptic curves over a prime field to imaginary quadratic orders. Nakaya [16] proposed analogous formulas for supersingular values of Hauptmoduli of Fricke groups. His Conjectures 5 and 6 concern the prime levels

$$\mathfrak{S} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}.$$

For $N \in \mathfrak{S}$, the curve $X_0(N)^+ = X_0(N)/\langle w_N\rangle$ has genus zero. Nakaya defines a squarefree polynomial $ss_p^{(N*)}$ from its distinct supersingular Hauptmodul values and denotes its number of linear factors over $\mathbb{F}_p$ by $L^{(N*)}(p)$. Conjecture 5 gives $L^{(N*)}(p)$, and Conjecture 6 gives the degree of $ss_p^{(N*)}$.

The point-counting identity behind the conjecture does not require genus zero. For a positive integer M and an exact divisor $d \mid M$, write

$$r_d(M) = \#\,\mathrm{Fix}\big(w_d \mid X_0(M)(\mathbb{C})\big),$$

whenever w_d is nonidentity. For a good-reduction quotient curve Y, let $\lambda_p(Y)$ denote the number of geometric supersingular points belonging to $Y(\mathbb{F}_p)$; precise definitions are given in Section 2.

Theorem 1.1. *Let $N \geq 1$, let $p \geq 5$ be prime with $p \nmid N$, and let W be a subgroup of the Atkin–Lehner group of $X_0(N)$. Put*

$$X = X_0(N)/W, \quad A = X_0(Np)/W, \quad B = X_0(Np)/\langle W, w_p \rangle.$$

The involution w_p induces a nonidentity involution $\bar{w}_p$ on A, and

$$\lambda_p(X) = 1 + g(A) - 2g(B) = \frac{1}{2} \# \operatorname{Fix}(\bar{w}_p \mid A(\mathbb{C})). \tag{1.1}$$

For the Fricke quotient, if $N > 1$, this specializes to

$$\lambda_p(X_0(N)^+) = \frac{r_p(Np) + r_{Np}(Np)}{4}. \tag{1.2}$$

For $N = 1$, the corresponding statement is $\lambda_p(X_0(1)) = r_p(p)/2$.

Here and below the genera and fixed-point counts are in characteristic zero, whereas λ_p counts points on the indicated good-reduction curve. The notation $X_0(1)^+ = X_0(1)$ is understood when discussing the level-one case.

Corollary 1.2 (Nakaya's Conjecture 5)**.** *Let $N \in \mathfrak{S} \setminus \{2\}$ and let $p \geq 5$ be prime with $p \neq N$. Then*

$$\begin{aligned} L^{(N*)}(p) = {} & \frac{1}{2}\left(1 + \left(\frac{-p}{N}\right)\right) L(p) \\ & + \frac{1}{8}\left\{2 + \left(1 - \left(\frac{-1}{Np}\right)\right)\left(2 + \left(\frac{-2}{Np}\right)\right)\right\} h(\mathbb{Q}(\sqrt{-Np})), \end{aligned} \tag{1.3}$$

where $L(p)$ is the number of rational classical supersingular j-invariants. The symbols with denominator Np are Jacobi symbols; they equal Nakaya's products of the two Legendre symbols. The quantity $h(\mathbb{Q}(\sqrt{-Np}))$ is a field class number.

Corollary 1.3 (Nakaya's Conjecture 6)**.** *Let $N \in \mathfrak{S} \setminus \{2\}$ and let $p \geq 5$ be prime with $p \neq N$. Let $d_N = \deg ss_N(X)$ be the number of supersingular j-invariants in characteristic N. Then*

$$\begin{aligned} \deg ss_p^{(N*)} = {} & \frac{(N+1)(p-1)}{24} + \frac{1}{8}\left(1 + \left(\frac{-1}{N}\right)\right)\left(1 - \left(\frac{-1}{p}\right)\right) \\ & + \frac{1}{6}\left(1 + \left(\frac{-3}{N}\right)\right)\left(1 - \left(\frac{-3}{p}\right)\right) + \frac{1}{2}\left(1 - \left(\frac{-N}{p}\right)\right) d_N. \end{aligned} \tag{1.4}$$

Here the symbols are Legendre symbols, $\left(\frac{-3}{3}\right) = 0$, and $d_3 = 1$ because $ss_3(X) = X$. The proof, in Section 6, replaces the rational count of Theorem 1.1 by the total count of supersingular points, which in genus zero is $1 + g(X_0(Np)/\langle w_N \rangle)$ by Lemma 3.1. Aricheta's theorem [2, Theorem 3.1], with the fact that every root lies in $\mathbb{F}_{p^2}$, already shows that the factors of $ss_p^{(N*)}$ over $\mathbb{F}_p$ that are not linear are exactly $g(X_0(Np)/\langle w_N, w_p \rangle)$ irreducible quadratics; the two corollaries supply the number $L^{(N*)}(p)$ of linear factors and the degree, and so complete the factorization (Corollary 6.3).

Relation to previous work. The geometric mechanism is classical. Deligne–Rapoport [6, VI, Theorem 6.9] describe the two-component special fibre of $X_0(Np)$ and its Frobenius gluing, and Ribet [21, Proposition 3.8(i), p. 446] states that w_p exchanges the components and acts as Frobenius on their intersection; Gross [8, Proposition 2.7(3)] gives the corresponding Brandt-matrix description. Ogg's reduction argument [19] is extended to Atkin–Lehner quotients by Aricheta [2, Theorem 3.1], using the quotient model of Furumoto–Hasegawa [7, Section 5];

Aricheta observes [2, Section 1] that his theorem settles Nakaya's characterization of the primes for which $ss_p^{(N*)}$ splits completely at $N = 5, 7$. For squarefree total level Np, which includes every case of Corollaries 1.2 and 1.3, Theorem 1.1 also follows from Beneish–Wen [3, Theorem 6.5]; see Section 3.1. For Conjecture 5, Nakaya proved $N = 2, 3$ [16, Theorem 4] and Morton proved $N = 5$ [12, 13]; for $N = 7$, Morton reduced it to the factor counts of the Hasse invariant of the Tate normal form E_7, of which he proved the linear and cubic counts [15, Theorem A] (see Remark 7.3). For Conjecture 6, Nakaya proved $N = 3$ [16], and Morton proved $N = 5$ [13, Section 7] and $N = 7$ [14, Section 3]. The contribution here is a uniform deduction of (1.3) and (1.4) at all fourteen levels from these ingredients, the versions at other prime levels (Corollary 5.2 and Proposition 6.2), and reproducible computations. No priority is claimed for the underlying geometric identities.

2 Supersingular quotient points and integral models

An exact divisor d of M satisfies $\gcd(d, M/d) = 1$. On a non-cuspidal point (E, C_M), with C_d the unique order-d subgroup of C_M, the Atkin–Lehner involution is

$$w_d(E, C_M) = (E/C_d, (C_M + E[d])/C_d).$$

In particular w_M replaces the cyclic isogeny by its dual. The involutions commute and satisfy $w_d w_e = w_{de/\gcd(d,e)^2}$; they generate the elementary abelian two-group indexed by exact divisors. We use their standard extensions to the compactified models. The modular curves throughout are coarse curves; a field-valued point therefore refers to the moduli orbit, without requiring a representative with all its structure defined over that field.

Let $p \geq 5$, $p \nmid N$, and W be as in Theorem 1.1. Let $\mathscr{X}_0(N)$ be the standard smooth proper coarse modular curve over $R = \mathbb{Z}_{(p)}$, and put $\mathscr{X} = \mathscr{X}_0(N)/W$. The order of W is a power of two and is invertible in R. The quotient is proper and flat over R. Indeed, locally its coordinate ring is the invariant subring of an R-flat ring, and the averaging idempotent makes this subring a direct summand. The same averaging argument shows that invariants commute with base change. Each geometric fibre of $\mathscr{X}$ is therefore a quotient of a smooth curve by a finite tame group. Such a quotient is normal and, over a perfect field, smooth. Consequently $\mathscr{X}$ is smooth over R.

Define $S_p(X)$ to be the image in $X(\overline{\mathbb{F}}_p)$ of the supersingular non-cuspidal points of $X_0(N)(\overline{\mathbb{F}}_p)$. A point of $X_0(N)$ represents a pair (E, C_N) with C_N cyclic of order N. Isogenous elliptic curves have the same supersingularity status, so $S_p(X)$ is well-defined on the quotient. These are geometric points of the coarse curve, with no automorphism weights. Define once and for all

$$D = \#S_p(X), \qquad Q_p(X) = \#(S_p(X) \setminus X(\mathbb{F}_p)), \qquad \lambda_p(X) = \#(S_p(X) \cap X(\mathbb{F}_p)). \tag{2.1}$$

Thus $D = \lambda_p(X) + Q_p(X)$.

These definitions agree with Aricheta's. In Section 3.1 of [2], he defines $Q_p(X)$ as "the number of supersingular points of X modulo p that are not defined over $\mathbb{F}_p$." The curve X there is already the quotient, not its covering modular curve. In particular, the counts refer to geometric points of the coarse quotient. This convention also follows directly from the two-copy quotient model used in his proof.

For clarity we restate the full statement of Aricheta's Theorem 3.1, with its notation expanded. Take any positive integer N, any exact divisors $e, f, \ldots$ of N, and $X = X_0(N) + e, f, \ldots$, the quotient by the group they generate. For any prime p not dividing N, put $X^p = X_0(Np) + p, e, f, \ldots$, with all products of these involutions included in the quotient group. Then

$$\tfrac{1}{2} Q_p(X) = g(X^p) - g(X). \tag{2.2}$$

In particular, all supersingular points on X are rational precisely when these genera are equal. Aricheta's statement permits primes smaller than 5; our proof uses only the tame range $p \geq 5$. Aricheta's proof is brief: it takes the two-copy model of $X_0(Np) + e, f, \ldots$ modulo p from [7, Section 5], notes that w_p exchanges the components and acts as Frobenius on their intersection, and follows Ogg's genus count [19]. Lemma 3.2 below reproves (2.2) for the quotients used here, supplying the flatness, base-change and local-ring details, as Lemma 3.1 does for the total count.

There is a relevant distinction between rational points upstairs and downstairs. If P is a supersingular point of $X_0(N)(\overline{\mathbb{F}}_p)$ with $P^{(p)} = w_N(P) \neq P$, its image x in $X_0(N)^+$ satisfies $x^{(p)} = x$. It contributes one to $\lambda_p(X_0(N)^+)$ and zero to $Q_p(X_0(N)^+)$. Requiring a rational lift would give an incorrect count. For an arbitrary W, the analogous condition is that the W-orbit of P is Frobenius-stable.

3 The geometric fixed-point lemma

We give the flat-family argument explicitly to fix the point-counting convention. The two shapes of special fibre that occur, two copies of $X_{\overline{\mathbb{F}}_p}$ meeting at the supersingular points and a single copy with self-crossings, are those described by Furumoto–Hasegawa [7, Section 5, Figures 1 and 2]; the lemmas below supply the node counts and the arithmetic genera.

Lemma 3.1. *With X, A as in Theorem 1.1 and D as in (2.1),*

$$g(A) = 2g(X) + D - 1. \tag{3.1}$$

Proof. Let $\mathscr{C}$ be the proper flat Deligne–Rapoport model of $X_0(Np)$ over R. Its geometric special fibre is the union of two copies of $X_0(N)_{\overline{\mathbb{F}}_p}$, with supersingular points glued using Frobenius. The coarse-curve statement, including the completed local form at an intersection, is [6, VI, Theorem 6.9(ii)–(iii), Corollary 6.10]. The modular crossings description is also treated by the Crossings Theorem of Katz–Mazur [11, Theorem 13.1.3, p. 390]; a further accessible formulation is [17, Theorem 2.7].

The quotient $\mathscr{A} = \mathscr{C}/W$ is proper and flat. Since $|W|$ is a unit, formation of its invariant rings commutes with reduction modulo p. Thus $\mathscr{A}_{\overline{\mathbb{F}}_p} = \mathscr{C}_{\overline{\mathbb{F}}_p}/W$. Every element of W acts on prime-to-p level structure and preserves each Deligne–Rapoport component. The normalized components of the quotient are consequently two copies of $X_{\overline{\mathbb{F}}_p}$.

At a node, write its completed special-fibre local ring as

$$\overline{\mathbb{F}}_p[[x, y]]/(xy) = \overline{\mathbb{F}}_p[[x]] \times_{\overline{\mathbb{F}}_p} \overline{\mathbb{F}}_p[[y]].$$

A stabilizer $H \subseteq W$ preserves the branches. Its invariant ring is

$$\overline{\mathbb{F}}_p[[x]]^H \times_{\overline{\mathbb{F}}_p} \overline{\mathbb{F}}_p[[y]]^H \simeq \overline{\mathbb{F}}_p[[u]] \times_{\overline{\mathbb{F}}_p} \overline{\mathbb{F}}_p[[v]] \simeq \overline{\mathbb{F}}_p[[u, v]]/(uv).$$

Here the invariants on each formal branch form a complete discrete valuation ring with residue field $\overline{\mathbb{F}}_p$. Therefore each orbit of intersection points gives exactly one ordinary double point. There are D such orbits, the very same supersingular points used in (2.1).

The fibres are reduced and connected. The Euler characteristic of the structure sheaf is constant in this proper flat family. On the special fibre, the normalization sequence gives

$$\chi(\mathcal{O}_{\mathscr{A}_{\overline{\mathbb{F}}_p}}) = 2\chi(\mathcal{O}_{X_{\overline{\mathbb{F}}_p}}) - D = 2(1 - g(X)) - D.$$

Its arithmetic genus is therefore $2g(X) + D - 1$, which equals the genus of the generic fibre A. □

The next lemma is Aricheta's identity (2.2) for the quotients used here. Its proof follows Aricheta's sketch and Ogg's argument [19], with the details of Lemma 3.1.

Lemma 3.2. *With X and B as in Theorem 1.1 and $D, \lambda_p(X)$ as in (2.1),*

$$g(B) = g(X) + \frac{D - \lambda_p(X)}{2}. \tag{3.2}$$

Proof. The involution w_p extends to $\mathscr{C}$, and in its geometric special fibre it exchanges the two components and acts on their intersection as Frobenius [21, Proposition 3.8(i)]. It commutes with W, so it induces an involution $\bar{w}_p$ of $\mathscr{A}$. The group $\langle W, w_p\rangle$ has order a power of two, so, exactly as for $\mathscr{A}$, the quotient $\mathscr{B} = \mathscr{A}/\langle \bar{w}_p\rangle$ is proper and flat over R, its formation commutes with reduction, and its generic fibre is B. Thus $\mathscr{B}_{\overline{\mathbb{F}}_p} = \mathscr{A}_{\overline{\mathbb{F}}_p}/\langle \bar{w}_p\rangle$.

The two components of $\mathscr{A}_{\overline{\mathbb{F}}_p}$, each isomorphic to $X_{\overline{\mathbb{F}}_p}$, are exchanged by $\bar{w}_p$, so the normalization of $\mathscr{B}_{\overline{\mathbb{F}}_p}$ is a single copy of $X_{\overline{\mathbb{F}}_p}$, and away from the nodes $\mathscr{B}_{\overline{\mathbb{F}}_p}$ is smooth. The D nodes of $\mathscr{A}_{\overline{\mathbb{F}}_p}$ correspond to the points of $S_p(X)$, and $\bar{w}_p$ permutes them as Frobenius does. A node fixed by $\bar{w}_p$ corresponds to a point of $S_p(X) \cap X(\mathbb{F}_p)$; there are $\lambda_p(X)$ of these. At such a node $\bar{w}_p$ exchanges the branches, so it acts on $\overline{\mathbb{F}}_p[[x]] \times_{\overline{\mathbb{F}}_p} \overline{\mathbb{F}}_p[[y]]$ by $(f, g) \mapsto (\alpha(g), \alpha^{-1}(f))$ for an isomorphism $\alpha\colon \overline{\mathbb{F}}_p[[y]] \to \overline{\mathbb{F}}_p[[x]]$. The invariants are the pairs $(f, \alpha^{-1}(f))$, a ring isomorphic to $\overline{\mathbb{F}}_p[[x]]$, so the image of the node is a smooth point of $\mathscr{B}_{\overline{\mathbb{F}}_p}$. The remaining $D - \lambda_p(X) = Q_p(X)$ nodes are exchanged in pairs, and $\bar{w}_p$ maps a neighbourhood of one isomorphically onto a neighbourhood of the other; each pair therefore gives one ordinary double point of $\mathscr{B}_{\overline{\mathbb{F}}_p}$, whose two branches lie on the same component of the normalization. Hence $\mathscr{B}_{\overline{\mathbb{F}}_p}$ is reduced and connected, with normalization $X_{\overline{\mathbb{F}}_p}$ and $Q_p(X)/2$ nodes, and the normalization sequence gives

$$\chi(\mathcal{O}_{\mathscr{B}_{\overline{\mathbb{F}}_p}}) = \chi(\mathcal{O}_{X_{\overline{\mathbb{F}}_p}}) - \tfrac{1}{2}Q_p(X).$$

By constancy of $\chi(\mathcal{O})$ in the flat family, $g(B) = g(X) + \frac{1}{2}Q_p(X)$. □

Lemma 3.3. *The first equality and the fixed-point equality in (1.1) hold.*

Proof. By (3.1), $D = 1 + g(A) - 2g(X)$, and by (3.2), $D - \lambda_p(X) = 2g(B) - 2g(X)$. Subtracting gives

$$\lambda_p(X) = D - Q_p(X) = 1 + g(A) - 2g(B).$$

The involution $\bar{w}_p$ is nonidentity: in the quotient model it exchanges the two components, whereas W preserves them. As $B = A/\langle \bar{w}_p\rangle$, Riemann–Hurwitz for this single involution gives

$$2g(A) - 2 = 2(2g(B) - 2) + \#\operatorname{Fix}(\bar{w}_p \mid A(\mathbb{C})).$$

Rearranging proves the assertion. □

In the proof of Lemma 3.2, a quotient point fixed by Frobenius is a node fixed by $\bar{w}_p$, whether or not it has a rational lift. Thus the points with non-rational lifts described in Section 2 are counted correctly.

Lemma 3.4. *If $N > 1$, then*

$$\#\operatorname{Fix}(\bar{w}_p \mid X_0(Np)/\langle w_N\rangle) = \tfrac{1}{2}(r_p(Np) + r_{Np}(Np)).$$

Proof. Put $C = X_0(Np)$, $u = w_N$ and $v = w_p$. The involutions are independent and commute, so generate a Klein four-group, with $uv = w_{Np}$. Independence may be seen from their action on the p-part and prime-to-p part of cusp labels.

No point of $C(\mathbb{C})$ is fixed by both u and v. The stabilizer of a point in a finite automorphism group of a compact Riemann surface injects, by its derivative, into $\mathbb{C}^*$, and is therefore cyclic.

Injectivity follows because a finite-order automorphism with derivative one is the identity in characteristic zero. A Klein four-group cannot be such a stabilizer.

A point $[P]$ of $C/\langle u\rangle$ is fixed by $\bar{v}$ exactly when $vP = P$ or $vP = uP$. Thus its preimage lies in $\mathrm{Fix}(v) \cup \mathrm{Fix}(uv)$. These sets are disjoint by the preceding paragraph. Also u acts freely on each, since a common fixed point would be fixed by the whole Klein group. Every resulting quotient point is therefore counted twice upstairs. This proves the formula. □

By Riemann–Hurwitz for the double covers of C by its three quotients of degree two and for $C/\langle u\rangle \to C/\langle u, v\rangle$, Lemma 3.4 is equivalent to the classical genus relation

$$g(C) + 2g(C/\langle u, v\rangle) = g(C/\langle u\rangle) + g(C/\langle v\rangle) + g(C/\langle uv\rangle)$$

for a Klein four-group, the case of Accola's formula [1, Theorem 1, Corollary, eq. (11), p. 600] for a group partitioned into three subgroups of order two; Accola's proof also rests on the cyclicity of point stabilizers.

Lemmas 3.3 and 3.4 prove Theorem 1.1 for $N > 1$. When $N = 1$, W is trivial, $A = X_0(p)$, and (1.1) already gives $r_p(p)/2$. No Klein-group argument is used in that case.

3.1 Relation to earlier quotient-model arguments

The comparison of genera in Lemmas 3.1–3.3 is the argument of Ogg [19] and Aricheta [2, Theorem 3.1], applied to both quotients A and B. Furumoto and Hasegawa [7, Section 5] describe the special fibres of the quotients of $X_0(pM)$, $p \nmid M$, by Atkin–Lehner subgroups, without node counts or genus formulas. Xue [23] constructs the minimal resolution over $\mathbb{Z}[1/6]$ of a single-involution quotient $X_0(N)/\langle w_M\rangle$, under conditions on the powers of 2 and 3 dividing N. In [23, Section 7.1, Theorem 2] he uses the quotient of $X_0(p)$ by w_p to prove Deuring's formula for the number of $\mathbb{F}_p$-rational supersingular j-invariants. The special fibre of that quotient is a rational curve with one node for each conjugate pair of non-rational supersingular j-invariants, and his proof compares its arithmetic genus with the genus of $X_0(p)$ from [6, VI, Theorem 6.9], using Riemann–Hurwitz and Ogg's fixed-point count [18, p. 454]. This is the case $N = 1$ of Theorem 1.1, proved by the same route.

Beneish and Wen [3] study squarefree total levels divisible by a prime $p \geq 5$. Their Theorem 6.5, stated as Theorem A in its introduction, describes both a quotient by an Atkin–Lehner subgroup preserving the two components and a further quotient containing a component-exchanging involution. Apply these two parts at total level Np, first with $H = W$ and then with $H = \langle W, w_p\rangle$. Their counts are expressed through the numbers $F(d)$ of w_d-fixed supersingular points on the special fibre. Their two-component intersection count $|W|^{-1}\sum_{w_d\in W} F(d)$ is D by Burnside's lemma. Since w_{dp} acts on supersingular points as w_d composed with Frobenius, the same lemma shows that $|W|^{-1}\sum_{w_d\in W} F(dp) = \lambda_p(X)$, so their self-crossing count $(2|W|)^{-1}\sum_{w_d\in W}(F(d) - F(dp))$ is $(D - \lambda_p(X))/2 = Q_p(X)/2$. Their genus formulas therefore give (1.1); single-involution Riemann–Hurwitz and Lemma 3.4 give (1.2). Hence their theorem already implies the fixed-point identity in the squarefree case. This deduction does not use their Proposition 3.16 or its additional hypotheses in certain ramified cases. The version above, for arbitrary N, uses the older general model instead. Beneish and Wen also record the characteristic-zero genus formula through Ogg's fixed-point counts [3, Remark 6.6]; they do not discuss Nakaya's supersingular polynomials.

The regular models constructed in [23] and [3] are not needed for the genus comparisons above. The arithmetic genus of the special fibre of a proper flat curve equals the genus of its generic fibre whether or not the total space is regular, and for $p \geq 5$ the tame quotients commute with reduction (Section 2). Together with the Deligne–Rapoport model and the action of w_p on it, these are the only inputs, and none of them requires N to be squarefree.

By the proof of Lemma 3.2, the special fibre of $\mathscr{B}$ is irreducible, with normalization $X_{\overline{\mathbb{F}}_p}$ and $Q_p(X)/2$ nodes. By the criterion of Padurariu, Park and Voight [20, Corollary 3.2], B therefore

has good reduction at p if and only if $Q_p(X) = 0$, that is, if and only if every supersingular point of X is $\mathbb{F}_p$-rational. Primes of good reduction dividing the level of full Atkin–Lehner quotients of squarefree level are classified in [20], and Beneish and Wen compare that work with their genus formula [3, Remark 6.8].

4 Characteristic-zero fixed points and class numbers

For a negative quadratic discriminant Δ, let $h_{\mathrm{ord}}(\Delta)$ denote the proper ideal class number of the order of discriminant Δ. It is distinct from the field class number when the order is nonmaximal.

We record the full squarefree-level specialization of the classical fixed-point formula before applying it. Let M be squarefree, $Q \mid M$, and $Q \geq 5$. For each odd prime $\ell \mid M/Q$, put

$$c_1(\ell) = c_2(\ell) = 1 + \left(\frac{-Q}{\ell}\right).$$

If $2 \mid M/Q$, then Q is odd and

$$c_1(2) = \begin{cases} 1, & Q \equiv 1 \pmod 4, \\ 2, & Q \equiv 3 \pmod 4, \end{cases} \qquad c_2(2) = 1 + \left(\frac{-Q}{2}\right) \quad (Q \equiv 3 \pmod 4).$$

The formula is

$$\begin{aligned} r_Q(M) = h_{\mathrm{ord}}(-4Q) \prod_{\ell \mid M/Q} c_1(\ell) \\ + \begin{cases} h_{\mathrm{ord}}(-Q) \displaystyle\prod_{\ell \mid M/Q} c_2(\ell), & Q \equiv 3 \pmod 4, \\ 0, & Q \not\equiv 3 \pmod 4. \end{cases} \end{aligned} \tag{4.1}$$

Ogg gives the odd-complement formula on p. 454 of [18], for $Q > 3$ and M/Q odd, with class numbers of primitive forms, that is, of orders; his local factor $1 + \left(\frac{-4Q}{\ell}\right)$ equals $c_1(\ell)$ because 4 is a square. The full formula, including the factors at 2 and the exceptional small divisors, is stated in [7, Remark 2, pp. 108–109] and [10, Proposition 2.1]. At squarefree level the cusps are labelled by divisors of M, and w_Q toggles the primes dividing Q in that label. Consequently a nonidentity w_Q fixes no cusp, in agreement with [18, Proposition 3]. Formula (4.1) is an unweighted count of geometric fixed points.

For an odd squarefree integer $d > 3$, define

$$V(d) = \begin{cases} h_{\mathrm{ord}}(-4d), & d \equiv 1 \pmod 4, \\ h_{\mathrm{ord}}(-4d) + h_{\mathrm{ord}}(-d), & d \equiv 3 \pmod 4. \end{cases}$$

Now let N be any odd prime distinct from $p \geq 5$. For $Q = p$, the only complementary prime in (4.1) is N. It does not divide either discriminant: $N \nmid 4p$. At an odd complementary prime the factors for the two possible orders coincide, since multiplication of the discriminant by 4 does not change its Legendre symbol. Thus

$$\begin{aligned} r_p(Np) &= \left(1 + \left(\frac{-p}{N}\right)\right) \begin{cases} h_{\mathrm{ord}}(-4p), & p \equiv 1 \pmod 4, \\ h_{\mathrm{ord}}(-4p) + h_{\mathrm{ord}}(-p), & p \equiv 3 \pmod 4, \end{cases} \\ &= \left(1 + \left(\frac{-p}{N}\right)\right) V(p). \end{aligned} \tag{4.2}$$

For $Q = Np$, the complementary product is empty, so

$$r_{Np}(Np) = V(Np). \tag{4.3}$$

Both divisors exceed 3, even if $N = 3$. No exceptional w_2 or w_3 term enters these two counts.

Deuring's formula, with field class numbers, is

$$L(p) = \begin{cases} \frac{1}{2}h(\mathbb{Q}(\sqrt{-p})), & p \equiv 1 \pmod 4, \\ 2h(\mathbb{Q}(\sqrt{-p})), & p \equiv 3 \pmod 8, \\ h(\mathbb{Q}(\sqrt{-p})), & p \equiv 7 \pmod 8. \end{cases} \tag{4.4}$$

See also the statement preceding [13, Theorem 1.2]. In particular $V(p) = 2L(p)$. When $d \equiv 1 \pmod 4$, the discriminant $-4d$ is fundamental. When $d \equiv 3 \pmod 4$, $-d$ is fundamental and the conductor-two class-number formula gives

$$h_{\mathrm{ord}}(-4d) = \left(2 - \left(\frac{-d}{2}\right)\right) h_{\mathrm{ord}}(-d).$$

The unit index is one because $d > 3$; the only imaginary quadratic fields with additional units would correspond to $d = 1$ or 3. It follows that

$$\frac{V(d)}{4} = \begin{cases} \frac{1}{4}h(\mathbb{Q}(\sqrt{-d})), & d \equiv 1 \pmod 4, \\ h(\mathbb{Q}(\sqrt{-d})), & d \equiv 3 \pmod 8, \\ \frac{1}{2}h(\mathbb{Q}(\sqrt{-d})), & d \equiv 7 \pmod 8. \end{cases} \tag{4.5}$$

Checking the four odd residue classes modulo 8 identifies this coefficient with

$$\frac{1}{8}\left\{2 + \left(1 - \left(\frac{-1}{d}\right)\right)\left(2 + \left(\frac{-2}{d}\right)\right)\right\}. \tag{4.6}$$

5 Hauptmodul coordinates and Conjecture 5

For $N \in \mathfrak{S}$, write $T_{N+} = q^{-1} + O(q)$ for the normalized principal modulus of $\Gamma_0(N)^+$. This is the Conway–Norton McKay–Thompson function for the prime Fricke group. Nakaya's normalizations are

N	j_N^*	Constant coefficient
3	$T_{3+} + 42$	42
5	$T_{5+} + 16$	16
7	$T_{7+} + 9$	9
$11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71$	T_{N+}	0

These are Nakaya's normalizations [16, Section 4.1]. At the eta-quotient levels $N = 2, 3, 5, 7, 13$, if $e = 24/(N-1)$, then

$$T_{N+} = \left(\frac{\eta(\tau)}{\eta(N\tau)}\right)^e + e + N^{e/2}\left(\frac{\eta(N\tau)}{\eta(\tau)}\right)^e. \tag{5.1}$$

For the auxiliary level 2, Nakaya takes $j_2^* = T_{2+} + 104$.

The integral Fourier expansions used by Nakaya in the proof of his Proposition 4 also follow from the moonshine realization [5, 4]: the normalized principal moduli are graded traces, and their rational coefficients are algebraic integers, hence integers. Borcherds's Theorem 1.1 identifies the traces with the Conway–Norton Hauptmoduli. The shifts displayed above preserve integrality.

Proposition 5.1. *For prime $N \in \mathfrak{S}$ and $p \geq 5$, $p \nmid N$, the function j_N^* gives an isomorphism $X_0(N)^+_{\mathbb{Z}_{(p)}} \simeq \mathbb{P}^1_{\mathbb{Z}_{(p)}}$. Under it, $S_p(X_0(N)^+)$ corresponds exactly to the roots of Nakaya's squarefree polynomial, and*

$$L^{(N*)}(p) = \lambda_p(X_0(N)^+), \qquad \deg ss_p^{(N*)} = D = \#S_p(X_0(N)^+). \tag{5.2}$$

Proof. The smoothness and base-change assertions for the tame quotient were proved in Section 2. Its fibres have genus zero and it has a rational cusp section c, the image of 0 and ∞. Relative Riemann–Roch gives a free rank-two module $H^0(\mathscr{X}, \mathcal{O}(c))$ and a coordinate t with principal part q^{-1} at c. The associated map identifies $\mathscr{X}$ with $\mathbb{P}^1_R$.

On the generic fibre, $j_N^* - t$ is constant, since both functions have exactly that simple pole and principal part. Its constant Fourier coefficient is in R, so this constant is in R. Therefore j_N^* is also an integral degree-one coordinate. This is the degree-one case of Fourier-expansion descent; the general q-expansion principle is [6, VII, Theorem 3.9]. One can instead apply that principle on a full-level cover: the expansions at both cusp orbits are integral by Fricke invariance, and translations introduce only roots of unity and prime-to-p changes of parameter. The relative Riemann–Roch argument above verifies the descent directly in the present situation.

Every finite Hauptmodul value now labels one point of the coarse quotient. That point is supersingular if and only if any representative elliptic curve is supersingular, since the representatives related by w_N are isogenous. A point has its value in $\mathbb{F}_p$ if and only if it belongs to $X(\mathbb{F}_p)$. The cusp corresponds to infinity and is excluded. Taking distinct values, as in Nakaya's definition, proves both statements in (5.2). □

Proof of Corollary 1.2. Combine (1.2), (5.2), (4.2), and (4.3):

$$L^{(N*)}(p) = \frac{1}{4}\left(1 + \left(\frac{-p}{N}\right)\right) V(p) + \frac{1}{4} V(Np).$$

Substitute $V(p) = 2L(p)$ and use (4.5)–(4.6) with $d = Np$. This is exactly (1.3). □

Only Proposition 5.1 uses the hypothesis $N \in \mathfrak{S}$. Theorem 1.1 holds for every $N > 1$, and Section 4 uses only that N is an odd prime different from p. The same computation therefore gives the count of rational supersingular points at every odd prime level, where $X_0(N)^+$ may have positive genus and no single polynomial carries the count.

Corollary 5.2 (All odd prime levels)**.** *Let N be an odd prime and let $p \geq 5$ be a prime with $p \neq N$. Then*

$$\lambda_p(X_0(N)^+) = \frac{1}{2}\left(1 + \left(\frac{-p}{N}\right)\right) L(p) + \frac{1}{8}\left\{2 + \left(1 - \left(\frac{-1}{Np}\right)\right)\left(2 + \left(\frac{-2}{Np}\right)\right)\right\} h(\mathbb{Q}(\sqrt{-Np})). \tag{5.3}$$

For $N \in \mathfrak{S} \setminus \{2\}$ this is Corollary 1.2.

Proof. Combine (1.2), (4.2) and (4.3), then substitute $V(p) = 2L(p)$ and (4.5)–(4.6) with $d = Np$, exactly as in the proof of Corollary 1.2, omitting (5.2). □

Remark 5.3 (Composite levels). Theorem 1.1 holds for every positive integer N prime to p and every Atkin–Lehner subgroup W; it does not require N to be squarefree. For the Fricke quotient of a squarefree composite N, inserting the squarefree-level formula (4.1) with $Q = p$ and $Q = Np$ into (1.2) gives class-number expressions for $\lambda_p(X_0(N)^+)$ with one local factor for each prime dividing N; both values of Q are at least 5. For non-squarefree N, explicit expressions require the full fixed-point formula at that level, such as [10, Proposition 2.1], with its hypotheses, its local factors at 2 and its orders of higher conductor. We do not record these cases here.

Appendix A.6 checks (5.3) by a direct count at sixteen prime levels outside $\mathfrak{S}$, using neither a Hauptmodul nor a genus or fixed-point formula.

6 The degree and Conjecture 6

Throughout this section N is an odd prime, $p \geq 5$ is a prime different from N, $X = X_0(N)^+$, $C = X_0(Np)$ and $A = C/\langle w_N \rangle$, and $D = \#S_p(X)$ is the total number of supersingular points of $X_{\overline{\mathbb{F}}_p}$. Lemma 3.1 with $W = \langle w_N \rangle$ gives

$$D = 1 + g(A) - 2g(X). \tag{6.1}$$

The involution w_N of C is not the identity, so $C \to A$ has degree two and is ramified exactly at the $r_N(Np)$ fixed points; no cusp is fixed [18, Proposition 3]. Riemann–Hurwitz gives $2g(C) - 2 = 2(2g(A) - 2) + r_N(Np)$, and hence

$$D = \frac{g(C)+3}{2} - \frac{r_N(Np)}{4} - 2g(X). \tag{6.2}$$

6.1 Genus and fixed points

Write

$$\alpha = \left(\frac{-1}{N}\right), \quad \beta = \left(\frac{-3}{N}\right), \quad \gamma = \left(\frac{-1}{p}\right), \quad \delta = \left(\frac{-3}{p}\right), \quad \kappa = \left(\frac{-N}{p}\right).$$

For a positive integer M, the genus of $X_0(M)$ is

$$g(X_0(M)) = 1 + \frac{\mu(M)}{12} - \frac{e_2(M)}{4} - \frac{e_3(M)}{3} - \frac{c(M)}{2},$$

where μ is the index of $\Gamma_0(M)$, e_2, e_3 count its elliptic points and c its cusps [9, Theorem 3.1.1 and Section 3.9]. For a prime N these are $\mu = N+1$, $e_2 = 1+\alpha$, $e_3 = 1+\beta$ and $c = 2$; at $N = 3$ the congruence $x^2 + x + 1 \equiv 0 \pmod 3$ has one root, matching $1 + \beta = 1$. These quantities are multiplicative over the distinct primes of a squarefree level, and $c(Np) = 4$. Hence

$$g_N := g(X_0(N)) = \frac{N+1}{12} - \frac{1+\alpha}{4} - \frac{1+\beta}{3}, \tag{6.3}$$

$$g(C) = \frac{(N+1)(p+1)}{12} - 1 - \frac{(1+\alpha)(1+\gamma)}{4} - \frac{(1+\beta)(1+\delta)}{3}. \tag{6.4}$$

Comparing (6.3) with the classical count of supersingular j-invariants [22, Theorem V.4.1(c)] gives $d_N = g_N + 1$ for every prime $N \geq 5$; for $N = 3$ both sides equal 1.

Lemma 6.1. *1. If $N \geq 5$, then $r_N(Np) = (1+\kappa)\, r_N(N)$.*

2. $r_3(3p) = 2(1+\delta)$ and $r_3(3) = 2$.

3. $r_N(N) = 2g_N + 2 - 4g(X)$. In particular $r_N(N) = 2d_N$ for $N \in \mathfrak{S} \setminus \{2\}$.

Proof. (1) Apply (4.1) with $Q = N$. For $M = N$ both products are empty. For $M = Np$ the only complementary prime is the odd prime p, and there $c_1(p) = c_2(p) = 1 + \left(\frac{-N}{p}\right) = 1 + \kappa$.

(2) Formula (4.1) assumes $Q \geq 5$. At $Q = 3$, Proposition 2.1 of [10] (see also [7, Remark 2]), applied at total level $M = 3p$ (written N there), so that $M/Q = p$, gives $r_3(3p) = c_1(p)\, h_{\mathrm{ord}}(-12) + \left(1 + \left(\frac{-3}{p}\right)\right)$ with $c_1(p) = 1 + \left(\frac{-3}{p}\right)$; the second summand is the exceptional term coming from the extra automorphisms of the curve with $j = 0$, and the $h_{\mathrm{ord}}(-Q)$ term is absent because it requires $Q \geq 4$. Since $h_{\mathrm{ord}}(-12) = 1$, this is $2(1+\delta)$. At total level $M = 3$ we have $M/Q = 1$, so both products are empty, giving $r_3(3) = h_{\mathrm{ord}}(-12) + 1 = 2$.

(3) Riemann–Hurwitz for the double cover $X_0(N) \to X$, ramified exactly at the $r_N(N)$ fixed points, gives $2g_N - 2 = 2(2g(X) - 2) + r_N(N)$. For $N \in \mathfrak{S} \setminus \{2\}$, $g(X) = 0$ and $g_N + 1 = d_N$. □

N	3	5	7	11	13	17	19	23	29	31	41	47	59	71
g_N	0	0	0	1	0	1	1	2	2	2	3	4	5	6
d_N	1	1	1	2	1	2	2	3	3	3	4	5	6	7
$r_N(N)$	2	2	2	4	2	4	4	6	6	6	8	10	12	14

6.2 Proof of Conjecture 6

Proof of Corollary 1.3. Let $N \in \mathfrak{S} \setminus \{2\}$. By Proposition 5.1, $\deg ss_p^{(N*)} = D$, and $g(X) = 0$. By Lemma 6.1, $r_N(Np) = 2d_N(1+\kappa)$ at every such level, including $N = 3$, where $1+\kappa = 1+\delta$. Substituting this and (6.4) into (6.2),

$$D = \frac{(N+1)(p+1)}{24} + 1 - \frac{(1+\alpha)(1+\gamma)}{8} - \frac{(1+\beta)(1+\delta)}{6} - \frac{d_N(1+\kappa)}{2}.$$

Subtracting the right-hand side of (1.4), every term involving γ, δ or κ cancels, and the difference is

$$\frac{N+1}{12} + 1 - \frac{1+\alpha}{4} - \frac{1+\beta}{3} - d_N = g_N + 1 - d_N = 0$$

by (6.3). This proves (1.4). □

At $N = 3$ the formula reads $\deg ss_p^{(3*)} = \frac{p-1}{6} + \frac{2}{3}\left(1 - \left(\frac{-3}{p}\right)\right)$, which is Nakaya's degree formula for that level [16]. No step requires $p > N$.

6.3 Other prime levels

Only Proposition 5.1 and the vanishing of $g(X)$ use $N \in \mathfrak{S}$. Without them, (6.2) and Lemma 6.1(3) still apply.

Proposition 6.2. *Let $N \geq 5$ and $p \geq 5$ be distinct primes. The number D of supersingular points of $X_0(N)^+$ in characteristic p equals the right-hand side of (1.4) minus $\left(1 - \left(\frac{-N}{p}\right)\right) g(X_0(N)^+)$.*

Proof. Write $g^+ = g(X)$. Now $r_N(Np) = (1+\kappa)(2d_N - 4g^+)$ by Lemma 6.1. Compared with the proof of Corollary 1.3, the term $-r_N(Np)/4$ in (6.2) gains $(1+\kappa)g^+$ and the term $-2g(X)$ contributes $-2g^+$, so D changes by $(1+\kappa)g^+ - 2g^+ = -(1-\kappa)g^+$. □

For $N \notin \mathfrak{S}$ there is no Hauptmodul, so D is not the degree of a single polynomial in this way. The correction vanishes exactly when $X_0(N)^+$ has genus zero or $\left(\frac{-N}{p}\right) = 1$. Appendix A.7 checks the proposition by a direct count at six levels where $X_0(N)^+$ has genus 1 or 2.

6.4 Factorization type

Corollary 6.3 (cf. [2, Theorem 3.1])**.** *Let $N \in \mathfrak{S} \setminus \{2\}$ and let $p \geq 5$ be prime with $p \neq N$. Over $\mathbb{F}_p$, the polynomial $ss_p^{(N*)}$ is a product of $L^{(N*)}(p)$ distinct linear factors and exactly $g(B)$ distinct irreducible quadratic factors, where $B = X_0(Np)/\langle w_N, w_p\rangle$.*

Proof. In the Deligne–Rapoport model, w_p exchanges the two components and acts on their intersection as Frobenius [21, Proposition 3.8(i)]. Since $w_p^2 = 1$, the square of Frobenius fixes every supersingular point, so every root of $ss_p^{(N*)}$ lies in $\mathbb{F}_{p^2}$ and the irreducible factors over $\mathbb{F}_p$ have degree one or two. By Proposition 5.1, the number of roots outside $\mathbb{F}_p$ is $D - \lambda_p(X) = Q_p(X)$, and Aricheta's theorem (2.2) gives $Q_p(X) = 2(g(B) - g(X)) = 2g(B)$, since $X^p = B$ and $g(X) = 0$. These roots come in $g(B)$ conjugate pairs. The count of quadratic factors is thus a direct consequence of Aricheta's theorem; Corollary 1.2 supplies the count of linear factors. □

7 Consistency checks and a worked example

Remark 7.1 (Level one). When $N = 1$, Theorem 1.1 gives $\lambda_p(X_0(1)) = r_p(p)/2 = V(p)/2 = L(p)$, recovering Deuring's formula by the route of Xue's proof [23, Theorem 2]. If one formally sets $r_{Np}(Np) = r_p(p)$ in (1.2), the same formula results; these are equal counts, not two independent involutions.

Remark 7.2 (Why level two has a different shape). Nakaya's Theorem 4 proves both the $N = 2$ and $N = 3$ explicit linear-factor formulas, as Morton also records in [14]. At level $2p$, (4.1) gives

$$r_p(2p) = \begin{cases} h_{\mathrm{ord}}(-4p), & p \equiv 1 \pmod 4, \\ 2h_{\mathrm{ord}}(-4p) + (1 + \left(\frac{-p}{2}\right))h_{\mathrm{ord}}(-p), & p \equiv 3 \pmod 4, \end{cases} \qquad r_{2p}(2p) = h_{\mathrm{ord}}(-8p).$$

Substitution in Theorem 1.1 and the conductor-two conversion give

$$L^{(2*)}(p) = \frac{1}{8}\left\{2 + \left(1 - \left(\frac{-1}{p}\right)\right)\left(4 + \left(\frac{-2}{p}\right)\right)\right\} h(\mathbb{Q}(\sqrt{-p})) + \tfrac{1}{4}h(\mathbb{Q}(\sqrt{-2p})). \tag{7.1}$$

This is Nakaya's formula. The separate exceptional small-divisor formula is

$$r_2(2p) = (1 + \left(\frac{-8}{p}\right))h_{\mathrm{ord}}(-8) + (1 + \left(\frac{-4}{p}\right)) = 2 + \left(\frac{-2}{p}\right) + \left(\frac{-1}{p}\right).$$

This term would appear in a genus calculation for the full Klein-group quotient, where it cancels; the proof of Theorem 1.1 does not use it. The different shape of (7.1) comes from the local factors at the complementary prime 2 in $r_p(2p)$. They are not the odd-prime factor $1 + (-p/N)$ used in (4.2). This explains the exclusion of 2 from the uniform conjecture.

The pair $N = 11$, $p = 13$. Here $j_{11}^* = T_{11+}$ has constant coefficient zero. The classical supersingular polynomial is $ss_{13}(X) = X - 5$. Using the Hauptmodul relation constructed in Appendix A, its distinct supersingular values give

$$\begin{aligned} ss_{13}^{(11*)}(T) &= T^8 + T^7 + 5T^6 + 5T^5 + T^4 + 4T^3 + 2T^2 + 9T + 11 \\ &= (T+2)(T+4)(T+5)(T+6)(T+11)(T+12)(T^2+2) \end{aligned}$$

in $\mathbb{F}_{13}[T]$. The quadratic factor is irreducible, so there are six linear factors. On the class-number side,

$$L(13) = 1, \quad \left(\frac{-13}{11}\right) = 1, \quad \left(\frac{-1}{143}\right) = -1, \quad \left(\frac{-2}{143}\right) = -1, \quad h(\mathbb{Q}(\sqrt{-143})) = 10.$$

The two summands of (1.3) are consequently

$$\tfrac{1}{2}(1+1)\cdot 1 = 1, \qquad \tfrac{1}{8}\{2 + (1+1)(2-1)\}\cdot 10 = 5.$$

Their sum is 6. Independently, enumeration of primitive reduced forms gives $h_{\mathrm{ord}}(-52) = 2$, $h_{\mathrm{ord}}(-143) = 10$, and $h_{\mathrm{ord}}(-572) = 10$. Therefore

$$r_{13}(143) = 4, \qquad r_{143}(143) = 20, \qquad (4 + 20)/4 = 6.$$

A further modular-form check gives the same fixed-point counts from $r_d(M) = 2 - 2\,\mathrm{tr}(w_d \mid S_2(\Gamma_0(M)))$.

For Conjecture 6 at the same pair, $g_{11} = 1$, $d_{11} = 2$, $\alpha = \beta = -1$ and $\kappa = \left(\frac{-11}{13}\right) = -1$, so (1.4) gives $6 + 0 + 0 + 2 = 8$. On the geometric side, $g(X_0(143)) = 13$ and $r_{11}(143) = (1 + \kappa)\, r_{11}(11) = 0$, so (6.2) gives $D = (13+3)/2 = 8$, the degree of the polynomial above. Its two non-rational roots form one conjugate pair, in agreement with Corollary 6.3: $B = X_0(143)/\langle w_{11}, w_{13}\rangle$ has genus 1.

Remark 7.3 (What the count does not determine). Morton's Hasse-invariant factor-count results imply the relevant Fricke linear-factor counts, but this implication cannot in general be reversed: the count is an aggregate of several factor types. In [15, Section 3, pp. 347–353] Morton shows that his Conjectures 1–3 on these factor counts [15, Section 2, pp. 345–346] imply Conjecture 5 for $N = 7$; his Theorem A [15, p. 342], proved in [15, Section 6], establishes Conjecture 1, which concerns the linear and cubic factors. Corollary 1.2 gives the case $N = 7$ unconditionally, but it does not by itself prove his finer quadratic and sextic assertions for E_7, which are his Conjectures 2 and 3; the abstract of [15] states them as conjectural. Complete splitting of a genus-zero supersingular polynomial is equivalent to $\lambda_p = D$, hence to Aricheta's rationality condition; identifying the associated sporadic-prime lists is already part of Aricheta's results.

A Exact computational verification

The PARI/GP script `verify_nakaya.gp`, in the repository named in the code and data availability statement, requires PARI/GP only. It constructs the Hauptmoduli and modular relations internally, without imports or coefficient tables. It was executed with PARI/GP 2.15.4 on 26 October 2026. The default survey contains 60 primes for $N = 3$ and 59 primes for each of the other thirteen levels, giving 827 pairs. All three counts below agreed in every case. A separate run for $N = 2$ verified (7.1) for all 60 primes. Full polynomials, relations, a CSV, and the run log are in the same repository.

A.1 Overview of the surveys

Table 1 lists every reported survey. All scripts are in the directory `verification` of the repository named in the code and data availability statement; commands are run from that directory. "Reconstructed" means that the run rebuilds the Hauptmoduli and modular relations from spaces of modular forms; "stored" means that it reads the relations `relations/relation_N.gp` produced by the reconstructing run under PARI/GP 2.15.4; "none" means that it uses only the classical supersingular polynomial and the modular polynomial Φ_N. Agreement in these surveys checks the implementations and normalizations; it does not prove the conjectures.

Table 1: Surveys, commands and expected outputs.

Survey	Command	Hauptmoduli	PARI	Expected final output
Conjecture 5, $p \le 300$	`gp -fq -s 256000000 verify_nakaya.gp`	reconstructed	2.15.4	`ALL PASSED: 827 cases`
Level 2, $p \le 300$	`NK_MAXP=300; NK_LEVELS=[2]; read("verify_nakaya.gp")`	reconstructed	2.15.4	`ALL PASSED: 60 cases`
Conjecture 5, $p \le 2000$	`python run_extended.py 2000`	stored	2.17.2	`TOTAL 4201 pairs, mismatches: none`
Other prime levels, Corollary 5.2	`verify_general.gp`, with `VG_LEVELS` and `VG_MAXP` as in Appendix A.6	none	2.17.2	`ALL AGREE: 468 cases`; `ALL AGREE: 58 cases`
Conjecture 6, $p \le 1500$	`python c6_degrees.py 1500`	stored	2.17.2	Part 1: 827 pairs, Part 2: 2478 pairs, `mismatches: none`
Conjecture 6, second implementation	in `conjecture6_second_implementation`: `gp -fq -s 768000000 code/verify_conjecture6.gp`, then `code/verify_traces.gp`	reconstructed	2.15.4	`ALL PASSED: 883`; `TRACE PASSED: 39`
Conjecture 6 and Proposition 6.2, direct	`gp -fq -s 1000000000 c6_direct.gp`	none	2.17.2	(a) 167 pairs, (b) 87 pairs, (c) 12 primes, each with `mismatches []`
Worked example	`gp -fq trace_example.gp`	none	2.15.4	fixed-point counts 4 and 20

A.2 Construction of the principal moduli

For each prime level set

$$k = \operatorname{lcm}\left(2, \frac{24}{\gcd(N+1, 24)}\right), \quad m = \frac{k(N+1)}{24}, \quad F_N = (\eta(\tau)\eta(N\tau))^k.$$

The form F_N has weight k, trivial character, no zero in the upper half-plane, order m at each cusp, and Fricke eigenvalue $(-1)^{k/2}$. Compute the eigenspace with that eigenvalue in $M_k(\Gamma_0(N))$ using exact rational matrices. Division by F_N identifies this eigenspace with the functions on the genus-zero quotient having pole order at most m at its cusp. Its dimension is $m+1$.

If c_N is the constant coefficient specified in Section 5, the numerator $f_N = j_N^* F_N$ is uniquely characterized by the coefficients

$$[q^i]f_N = 0 \ (0 \le i < m-1), \quad [q^{m-1}]f_N = 1, \quad [q^m]f_N = c_N - k.$$

The last equality uses $F_N = q^m(1 - kq + O(q^2))$. Solving these linear conditions gives $j_N^* = f_N/F_N$ exactly. The code checks the dimension, leading coefficient, constant coefficient, and integrality of the retained Fourier coefficients.

Pole-part elimination then constructs the polynomials a_N, b_N satisfying

$$j(\tau) + j(N\tau) = a_N(j_N^*), \qquad j(\tau)j(N\tau) = b_N(j_N^*).$$

They have degrees N and $N+1$. Matching the entire principal part and constant coefficient certifies these identities: the difference is a holomorphic function on the compact quotient and vanishes at the cusp. Additional positive coefficients are checked as a safeguard. The corresponding quadratic is $R_N(X,T) = X^2 - a_N(T)X + b_N(T)$.

A.3 The three computations

For the direct polynomial computation use the classical supersingular polynomial. An explicit formula, also used in [13], is obtained by setting $n = \lfloor p/12 \rfloor$, $\delta = (1 - (-3/p))/2$, $\varepsilon = (1 - (-1/p))/2$ and taking the monic normalization of

$$X^\delta (X - 1728)^\varepsilon \sum_{j=0}^{n} \binom{2n+\varepsilon}{2j+\varepsilon} \binom{2n-2j}{n-j} (-432)^{n-j} (X - 1728)^j.$$

The desired polynomial is computed as the radical of

$$\operatorname{Res}_X(ss_p(X), R_N(X,T)).$$

The radical routine handles derivative-zero factors by extracting pth roots, so multiplicities divisible by p are not lost. The number of distinct rational roots is

$$\deg \gcd(ss_p^{(N*)}(T), T^p - T).$$

Separately, the program evaluates (1.3) with field discriminants and PARI's `qfbclassno`.

For the independent arithmetic fixed-point computation, it enumerates primitive positive definite reduced forms $[a,b,c]$ with $b^2 - 4ac = \Delta$, $|b| \le a \le c$, the usual boundary sign convention, and $\gcd(a,b,c) = 1$. At odd complementary primes it enumerates the solutions of $z^2 + Q = 0$ rather than inserting a precomputed Legendre symbol. The separate local rule at 2 is used for the supplementary run. This computes the characteristic-zero counts in (4.1) without calling `qfbclassno`. Independence here means separate arithmetic implementations; it is not a second proof of the fixed-point classification. Every computed polynomial is also checked to split over $\mathbb{F}_{p^2}$.

Table 2: Representative exact checks. The degree of the computed polynomial agrees with (1.4), and its number of rational roots with (1.3). The last two columns are independently enumerated fixed-point counts; their sum divided by four equals the number of rational roots.

N	p	$\deg ss_p^{(N*)}$	(1.4)	Direct roots	(1.3)	$r_p(Np)$	$r_{Np}(Np)$
3	13	2	2	2	2	0	8
5	13	4	4	2	2	0	8
7	13	5	5	3	3	4	8
11	13	8	8	6	6	4	20
13	17	10	10	6	6	8	16
17	13	9	9	5	5	4	16
19	13	12	12	4	4	4	12
23	13	12	12	8	8	0	32
29	13	15	15	5	5	4	16
31	13	19	19	3	3	4	8
41	13	25	25	3	3	0	12
47	13	29	29	11	11	4	40
59	13	36	36	12	12	4	44
71	13	43	43	11	11	4	40

A.4 Reproduction

Run the following command in a fresh directory containing the script:

```
gp -fq -s 256000000 verify_nakaya.gp
```

It writes `verification.csv`, `relation_N.gp`, and `polynomials_N.gp`, with one relation and polynomial file for each level. The final output must be `ALL PASSED: 827 cases`. To run the level-two supplement from an interactive session in another directory, use

```
NK_MAXP=300; NK_LEVELS=[2]; read("verify_nakaya.gp");
```

To load functions without running a survey set `NK_NO_RUN=1` before `read`. A direct trace check for the worked example is

```
mf=mfinit([143,2],1);
at=mfatkininit(mf,13); print(2-2*trace(at[2]/at[3]));
at=mfatkininit(mf,143); print(2-2*trace(at[2]/at[3]));
```

The output is 4 and 20.

A.5 Version note and extended check

Under PARI/GP 2.17.2 we observed that the Hauptmodul construction stops at $N = 13$ with the message `unexpected Hauptmodul dimension`; levels $3, 5, 7, 11$ run unchanged. At $N = 13$ the construction uses `mf=mfinit([13,12],4)`, the full space $M_{12}(\Gamma_0(13))$ of dimension 15, and `at=mfatkininit(mf,13)`, whose normalizing constant `at[3]` is 1. In that environment the matrix `at[2]` does not satisfy `at[2]`$^2 = 1$, and its $+1$ eigenspace, the eigenvalue $(-1)^{k/2}$ required for $k = 12$, has dimension 7 rather than the expected $m + 1 = 8$, so the dimension check fails. We have not isolated the cause, and we do not claim a defect in PARI/GP; the same construction passed all checks under 2.15.4. Users of later versions can instead load the stored files `relation_N.gp` produced under 2.15.4, set `NK_NO_RUN=1`, and call `nk_polynomial`, `nk_rhs` and `nk_fixed` directly. Run in this way under PARI 2.17.2, using the stored relations, the three counts agreed for all 14 levels and every prime $5 \le p \le 2000$ with $p \neq N$, a total of 4201 pairs.

A.6 Direct count at prime levels outside $\mathfrak{S}$

The script `verify_general.gp` checks Corollary 5.2 without Hauptmoduli, genus formulas or the fixed-point formula (4.1). It uses only the classical supersingular polynomial and the modular polynomial $\Phi_N(x, y)$, computed by PARI's `polmodular`.

It restricts to primes $p \equiv 1 \pmod{12}$, so that $j = 0$ and $j = 1728$ are ordinary and every supersingular curve has automorphism group $\{\pm 1\}$. For each supersingular $j \in \mathbb{F}_{p^2}$, the roots j' of $\Phi_N(j, Y)$, counted with multiplicity m, give the $N+1$ points of $X_0(N)$ over j. The involution w_N sends the class (j, j') to (j', j), and Frobenius sends it to (j^p, j'^p). A w_N-orbit of classes that Frobenius does not preserve contains no rational point of $X_0(N)^+$, and a Frobenius-stable class with $m = 1$ is one rational point.

A Frobenius-stable class with $m > 1$ arises, for example, when $j, j' \in \mathbb{F}_p$ and a subgroup C and its conjugate $C^{(p)}$ have the same quotient. These are the points with non-rational lifts discussed in Section 2. Since $X_0(N)$ is the normalization of the plane curve $\Phi_N = 0$ and the map to the j-line is étale away from $j = 0, 1728$, the m points over (j, j') are m smooth branches with x as local parameter. Their tangent slopes s are the roots of the degree-m tangent cone. Frobenius sends slope s to s^p at (j^p, j'^p), and w_N sends it to $1/s$ at (j', j); this decides which points of $X_0(N)^+$ are rational. When the tangent cone has a repeated slope the prime is reported and skipped.

The resulting count was compared with (5.3) and with $(r_p(Np) + r_{Np}(Np))/4$ from (4.2)–(4.3). Run under PARI/GP 2.17.2:

Levels N	Range	Pairs	Mismatches
$11, 13, 17, 19, 23$ (in $\mathfrak{S}$)	$p \le 1300$	230	0
$37, 43, 53, 61, 67, 73$	$p \le 1300$	238	0
$79, 83, 89, 97, 101, 103, 107, 109, 113, 127$	$p \le 600$	58	0

In these runs 10,881 Frobenius-stable multiple points were resolved by tangent slopes. A further 237 pairs, all with $p < 4N$, had a repeated tangent slope and were skipped. Primes $p \not\equiv 1 \pmod{12}$ are not covered by this check; at levels in $\mathfrak{S}$ they are covered by the main survey. Finite agreement is evidence for the implementation and normalizations, not a substitute for the proof.

A.7 Checks of the degree formula

The degree checks below are independent of the proof in Section 6; like the checks above, they test the implementation and normalizations.

1. *Degrees from the main survey.* The column of polynomial degrees in `verification.csv` agrees with (1.4) for all 827 pairs with $p \le 300$. Fresh degrees computed from the stored relations under PARI/GP 2.17.2 agree for all 2478 pairs with $300 < p \le 1500$ at all fourteen levels (script `c6_degrees.py`).

2. *Second implementation.* A separate PARI/GP program, which rebuilds the Hauptmoduli and computes the resultant radicals independently, agreed with (1.4) under PARI/GP 2.15.4 for all 883 pairs tested: every prime $5 \le p \le 300$ and $p = 307, 401, 503, 1009$ at all fourteen levels, including 89 pairs with $p < N$. A trace computation of $g(A)$ and $r_N(Np)$ on the full space of weight-two cusp forms agreed in all 39 cases with $p = 5, 7, 13$.

3. *Direct count from Φ_N.* For $p \equiv 1 \pmod{12}$ the total count D was obtained from the modular polynomial alone, as in Appendix A.6: a w_N-orbit $\{(j, j'), (j', j)\}$ with $j \neq j'$ and multiplicity m contributes m points of $X_0(N)^+$, and the m branches at a loop (j, j) contribute the orbits of $s \mapsto 1/s$ on their tangent slopes. The count agreed with (1.4)

in all 167 pairs at $N = 11, \dots, 31$, $p \le 700$, and with Proposition 6.2 in all 87 pairs at $N = 37, 43, 53, 61, 67, 73$, $p \le 600$, where $X_0(N)^+$ has genus 1 or 2 (script `c6_direct.gp`). Pairs with a repeated tangent slope at a loop were skipped and reported (21 and 48 pairs).

4. *Level three.* Atkin–Lehner traces give $r_3(3p) = 2\big(1 + \left(\frac{-3}{p}\right)\big)$ for all primes $5 \le p \le 43$, as in Lemma 6.1(2).

Code and data availability

The PARI/GP and Python scripts, stored modular relations, computed polynomials, CSV files and logs described in Appendix A are available in the repository

`https://github.com/bbrhuft/nakaya-conjectures`

The version matching this manuscript should be cited by a tagged release or commit hash. No archival DOI has been assigned.

Declaration of generative AI assistance

OpenAI ChatGPT-6.1 Sol assisted with literature retrieval, development and checking of the mathematical exposition, including the first version of the degree argument, preparation and execution of verification code, and manuscript editing. Anthropic Claude Opus 5.5 assisted with checking the arguments and the cited statements of Aricheta, Ogg, Im–Jeon–Kim, Beneish–Wen and Morton against the original sources, the extended verification run, the direct-count verifiers of Appendices A.6 and A.7, merging the two parts of this article, and copy-editing. Google Gemini 4 Argon assisted with manuscript review and a literature survey. Authorial responsibility for the mathematical claims, references, computations, and final text rests with the author.

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Maynooth University Social Sciences Institute (MUSSI), Maynooth University, Maynooth, County Kildare, Ireland
Email address: David.Jordan@mu.ie